StockSense AI

A team walkthrough

The app, the stock calculation and the models

What everyone should understand

01 Read the forecast and explain its dates.

02 Calculate a stock order and explain the warning.

03 Explain what each model does.

04 Interpret the results and answer a lecturer’s question.

The manager’s stock decision

How many units should Sunrise Mini Market order?

The app estimates demand.

The manager enters stock and delivery details, then reviews the suggestion.

Our dataset and scope

1

fictional store

5

anonymous products

7

forecast days

913,000 source rows. 9,130 rows in our chosen scope.

Daily sales from 2013 to 2017

What one data row means

Date	Store	Item	Sales (units)
2013-01-01	1	1	13
2013-01-02	1	1	11
2013-01-03	1	1	14

13 means units sold on that date.

Stock levels, prices and supplier details require separate inputs.

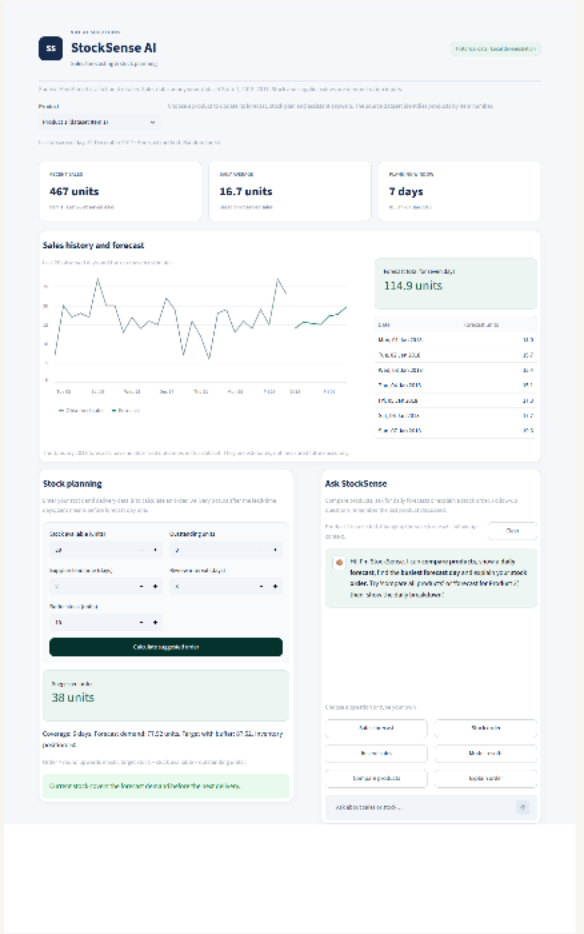
The system at a glance

01 Product selection and recent sales

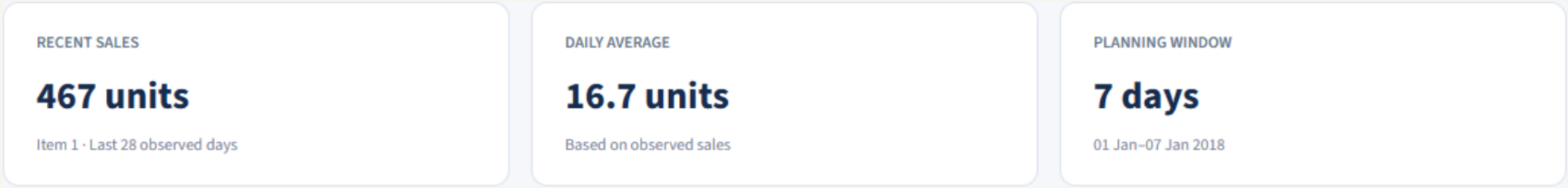
02 History and seven-day forecast

03 Stock planning and warnings

04 Questions and model results



Product selection and the summary



Recent sales: the last 28 observed days.

Daily average: recent total divided by 28.

The Product selector updates the view and resets old context.

History and forecast



Cutoff: 31 Dec 2017, the last observed day.

Horizon: seven forecast days, 1–7 Jan 2018.

The daily forecast in units

Date	Forecast units
Mon, 01 Jan 2018	14.0
Tue, 02 Jan 2018	15.7
Wed, 03 Jan 2018	15.4
Thu, 04 Jan 2018	15.1
Fri, 05 Jan 2018	17.3
Sat, 06 Jan 2018	17.7
Sun, 07 Jan 2018	19.6

114.9

forecast units for the week

Totals use unrounded estimates.
Orders round up to whole units.

The five stock inputs

Input	Plain-English meaning	Default
Stock available	Units physically in the shop now	50 units
Outstanding units	Units already ordered and expected	0 units
Supplier lead time	Forecast days before the next delivery	2 days
Review interval	Days between stock reviews	3 days
Buffer stock	Extra units above forecast demand	10 units

“Calculate suggested order” applies the entered values.

Coverage and delivery timing

2 lead days + 3 review days = 5 coverage days

Day	Mon 1	Tue 2	Wed 3	Thu 4	Fri 5	Sat 6	Sun 7
Forecast	14.0	15.7	15.4	15.1	17.3	17.7	19.6
Timing	Before delivery	Before delivery	Review coverage	Review coverage	Review coverage	Outside coverage	Outside coverage

Five-day demand = 77.52 units

Why the app suggests 38 units

77.52

five-day forecast demand

+ 10 buffer = 87.52 target

50 available + 0 outstanding = 50 position

38

units to order

87.52 − 50 = 37.52

Rounded up

The warning before delivery

Before the delivery	Units
First two days’ forecast demand	29.8
Stock physically available	5.0
Potential shortage	24.8

The new order is 83 units.

An order arriving later cannot cover earlier sales.

Checks that protect the calculation

01 Whole, non-negative units and days.

02 Lead time + review interval must be at most 7 days.

03 Enough stock can give an order of 0 units.

04 A new product needs its own submitted stock calculation.

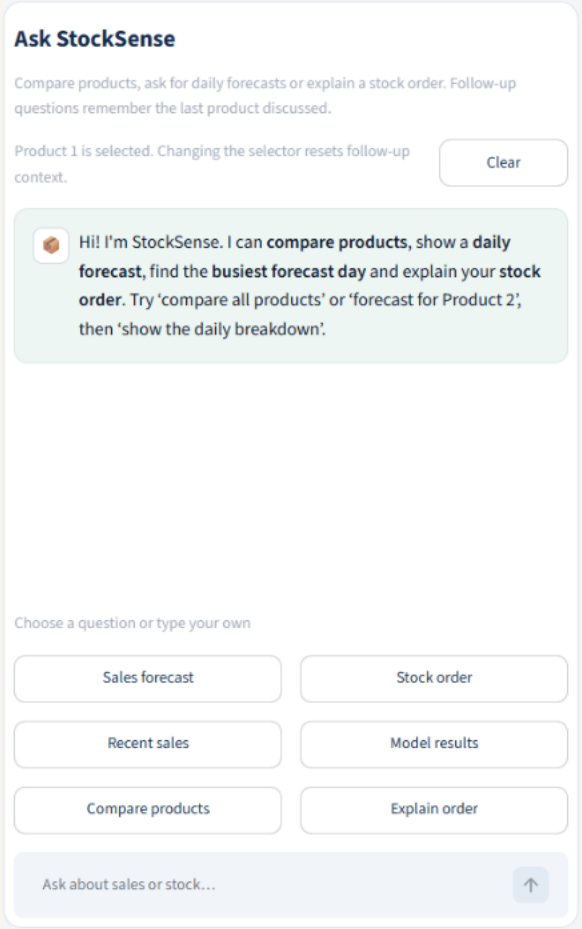
Ask StockSense

“Compare all products”

“Which day has the highest forecast?”

“What does buffer stock mean?”

“Explain my suggested order”



Follow-up questions and product focus

Question	What the assistant uses
“Forecast for Product 2”	Product 2: 310.6 forecast units
“And Product 3?”	Product 3: 178.7 forecast units
“Show the daily breakdown”	Product 3: seven dated estimates
“Why?” after calculating an order	The submitted order’s calculation

Clear resets the chat focus. The selector also resets it.

How the chatbot understands a question

01 Words become numeric features using TF-IDF.

02 Logistic regression predicts the request type.

03 Product rules and follow-ups identify the relevant data.

04 The answer uses actual forecast and stock results.

Useful answers and clear limits

Request	Where the answer comes from
Forecast or busiest day	The item’s seven forecast values
Recent sales	The item’s observed sales history
Order and “why?”	Its submitted stock plan
Model results	Saved validation and final-test metrics
A definition or help	Supported explanations and example questions

Intent confidence concerns the question category.

The three forecasting methods

Method	How it forecasts	Role
Weekly seasonal naive	Copies the same weekday from last week	Simple benchmark
Random forest	Averages predictions from many trees	Selected forecast
Small LSTM	Learns from a sequence of daily sales	Comparison model

The chatbot classifies questions. These methods forecast units.

Weekly seasonal naive

Day	Last week observed	Next week forecast
Monday	10 units	10 units
Tuesday	12 units	12 units
Wednesday	9 units	9 units
Thursday	11 units	11 units

Illustration: the same rule repeats for all seven weekdays.

Random forest

100

decision trees per product

Inputs: 28 past sales values, recent averages and known calendar dates.

The forest predicts all seven daily values together.

LSTM: Long Short-Term Memory

28

past daily values

16

LSTM memory units

7

daily outputs

One small neural network per product

Sales values only. Training scales the values before learning.

Training, validation and final testing

Period	Purpose
2013–2015	Initial model fitting
2016	Validation selects the method and settings
Through 2016	Refit with the selected settings
2017	Final test of the frozen choices

52 complete weeks per product in each evaluation year

MAE: average forecast mistake

Day	Observed	Forecast	Absolute error
Day 1	10	12	2
Day 2	20	17	3

(2 + 3) ÷ 2 = 2.5 units

Illustration. Smaller error is better.

RMSE and weekly-total MAE

Root Mean Squared Error

Larger mistakes receive more weight.

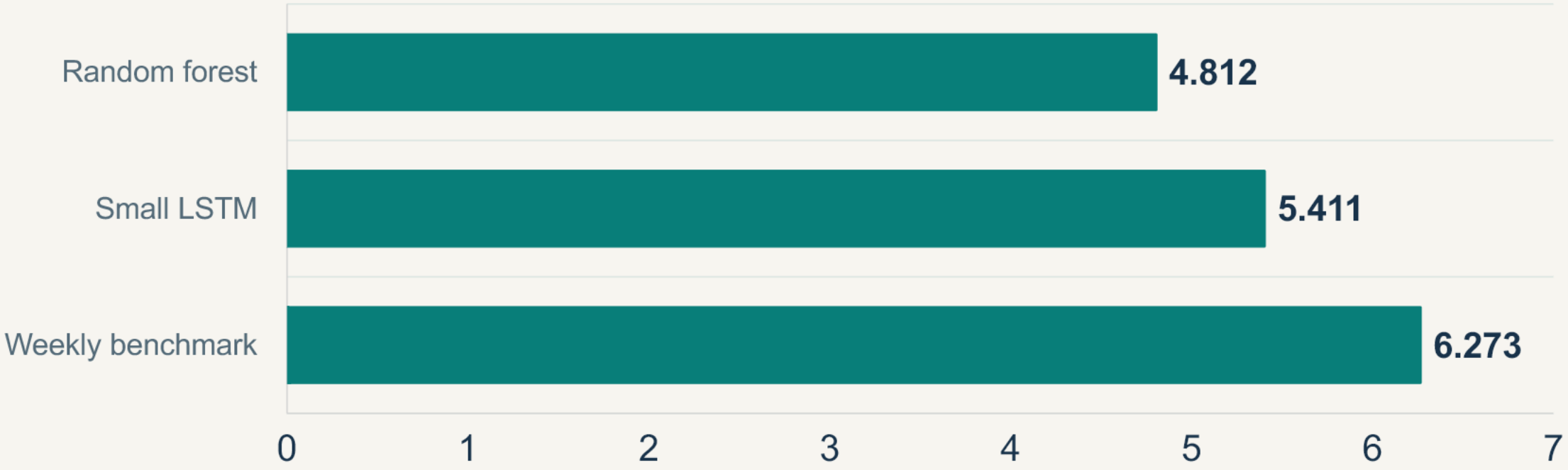
Previous slide's example: 2.55 units

Weekly-total MAE compares seven-day totals.

Two-day illustration	Actual	Forecast
Day 1	10	12
Day 2	20	18
Total	30	30

Daily MAE = 2 units. Total error = 0.

Random forest's daily test error



23.3% lower daily MAE than the benchmark

ALL ERRORS IN UNITS. SMALLER IS BETTER

The recorded model results

Period and method	Daily MAE	Daily RMSE	Weekly-total MAE
2016 Weekly benchmark	6.320	8.217	18.758
2016 Random forest	4.911	6.596	19.797
2016 Small LSTM	5.439	7.227	17.280
2017 Weekly benchmark	6.273	8.218	18.427
2017 Random forest	4.812	6.387	17.973
2017 Small LSTM	5.411	7.261	17.343

LSTM has the lowest weekly-total error. Daily MAE chose random forest.

What our evidence supports

01 A historical comparison of forecasting errors.

02 A stock suggestion from the values we enter.

03 A focused assistant that reads the system’s results.

04 Real savings and January 2018 accuracy remain unmeasured.

A live demonstration everyone can repeat

01 Select Product 1 and explain the dates.

02 Read its seven daily forecasts and total.

03 Calculate the default 38-unit order.

04 Change stock to 5 and explain the delivery warning.

05 Compare products and try a follow-up question.

06 Open the 2016 and 2017 model results.

Questions for the group

01 Why does the order use 77.52 units instead of 114.9?

02 Does a daily MAE of 4.812 mean 95.188% accuracy?

03 Why did we choose random forest when LSTM has a lower weekly error?

04 What does the chatbot need before suggesting another product’s order?

Model terms in plain English

Term	Meaning in this project
AI and machine learning	AI is the broad field. Machine learning learns from examples.
Regression	Predicting a number, such as daily units sold
Classification	Predicting a category, such as a question’s intent
Feature	An input, such as a past sales value or weekday
Target	The known value a model learns to predict
Lag	A previous sales value, such as yesterday’s sales
Ensemble	Several models combined, such as the forest’s trees
Deep learning	Learning with neural networks, such as our LSTM

Training and evaluation terms

Term	Plain-English meaning
Epoch and batch	Epoch: a training pass. Batch: a group for one update.
Optimizer and loss	The update method and the error it tries to reduce
Scaling	Putting training values onto a comparable numeric scale
Hyperparameter	A chosen setting, such as tree count or learning rate
Overfitting	Learning training detail that does not generalize well
Data leakage	Using future information when predicting earlier dates
Inference	Running a trained model to produce an answer
Confidence threshold	Minimum score before accepting a question category

Each member’s next contribution

One feature. One explanation.

Each member demonstrates something and answers a question.

The group records its actual decisions and follow-up work in GitHub.